

Design FMEA (qualitative, simulation-signal based) — VBF-T2R1FLASH-DFMEA-01

Qualitative caliber: severity/occurrence rated 3-level (H/M/L) from simulation deviation magnitude vs threshold; RPN = S x O qualitative matrix. Complete FMEA (process/supplier) N/A (beyond pure simulation boundary).

#	Failure mode	Failure cause	Simulation signal (detectability)	S	O	RPN	Design-side mitigation (implemented)
1	Negative electrode Li plating @4C fast charge	Transport/kinetics overpotential pushes anode potential < 0 V	anode_potential_v min = +12.3 mV (baseline -191.8 mV)	H	L	M	Small negative particles (2.5 um), high-sigma/high-t+ electrolyte, porosity 0.30
2	Thermal runaway risk (temp rise over red line)	Ohmic + reaction heat at 4C	T_max 329.2 K vs 333.15 K (baseline 354.3 K)	H	L	M	Active liquid cooling h=60 W/m2K + polarization reduction
3	SEI overgrowth -> capacity fade	Baseline SEI kinetics	SEI 262 nm @500cyc vs 550 (baseline 778 nm)	M	L	L	SEI-suppression coating bridge (k x0.002, artificial-SEI class)
4	Low-T capacity loss	Transport freeze-out at -20C	retention 99.4 % vs 90 % (baseline 99.4 %)	M	L	L	High-transport electrolyte (sigma/D/t+ overrides)
5	Insufficient energy density	Electrode loading/mass balance	425.4 Wh/kg vs 327.18 (baseline 400.8)	H	L	L	n/a needed (margin)
6	Electrolyte oxidative decomposition	High-voltage exposure	HOMO/IE-EA vs window: N/A (real_compute=false, no DFT endorsement)	M	L	L	Voltage window 4.2 V within NMC811 stability (lit.); true DFT endorsement not run (case meta)

Conclusion: highest-risk item (4C plating) mitigated in design (anode min +12 mV margin); all simulated signals within limits. Nail/overcharge/crush/process failure modes: N/A (beyond pure simulation boundary).